Exercise 1 (SQL Fundamentals)

```
1. SELECT *
   FROM employees;
```

id	First_name	last_name	department	Salary	hire_date	city
1	John	Doe	IT	55 000	2018-06-15	New York
2	Jane	Smith	HR	48000	2019-07-20	Chicago
3	Mike	Johnson	Finance	60000	2017-09-30	Los Angeles
4	Sarah	Brown	IT	53000	2021-03-25	New York
5	David	White	Marketing	52000	2016-04-10	San Francisco
6	Emily	Davis	IT	62000	2015-02-14	Chicago
7	Robert	Wilson	Finance	59000	2019-10-01	Houston
8	Jessica	Moore	HR	51000	2018-05-22	Los Angeles
9	Daniel	Clark	Marketing	53000	2022-06-01	Chicago
10	Laura	Hall	IT	50000	2020-08-10	San Francisco

```
2. SELECT DISTINCT department
   FROM employees;
```

department
IT
HR
Finance
Marketing

```
3. SELECT first_name,
          last_name,
          salary
   FROM employees
   ORDER BY salary DESC;
```

First_name	last_name	Salary
Emily	Davis	62000
Mike	Johnson	60000
Robert	Wilson	59000
John	Doe	55000
Sarah	Brown	53000
Daniel	Clark	53000
David	White	52000
Jessica	Moore	51000
Laura	Hall	50000
Jane	Smith	48000

4.
```
SELECT first_name,
       last-name,
       salary
FROM   employees
ORDER BY salary DESC
LIMIT 5;
```

id	first_name	last_name	salary
6	Emily	Davis	62 000
3	Mike	Johnson	60 000
7	Robert	Wilson	59000
1	John	Doe	55000
4	Sarah	Brown	53 000

5.
```
SELECT first-name,
       last-name,
       department
FROM   Employees
WHERE  department = 'IT';
```

first-name	last-name	department
John	Doe	IT
Sarah	Brown	IT
Emily	Davis	IT
Laura	Hall	IT

6. SELECT first-name,
   last-name,
   salary
   FROM employees
   WHERE department = 'Finance' AND salary > 58,000;

id	first-name	last-name	salary
6	Emily	Davis	62000
3	Mike	Johnson	60000
7	Robert	Wilson	59000

7. SELECT first-name,
   last-name,
   department
   FROM employees
   WHERE department= 'HR' OR department= 'Marketing';

first_name	last_name	department
Jane	Smith	HR
David	White	Marketing
Jessica	Moore	HR
Daniel	Clark	Marketing

8. SELECT first-name,
last-name,
department
FROM employees
WHERE department != 'IT';

id	first-name	last-name	department
2	Jane	Smith	HR
3	Mike	Johnson	Finance
5	David	White	Marketing
7	Robert	Wilson	Finance
8	Jessica	Moore	HR
9	Daniel	Clark	Marketing

9. SELECT first-name,
last-name,
department
FROM employees
WHERE department IN ('IT', 'HR', 'Finance');

First_name	last-name	department
John	Doe	IT
Jane	Smith	HR
Mike	Johnson	Finance
Sarah	Brown	IT
Emily	Davis	IT
Robert	Wilson	Finance
Jessica	Moore	HR
laura	Hall	IT

0.
```
SELECT First-name,
       last_name,
       department,
       salary,
       City
FROM   employees
WHERE  department = 'IT' AND salary > 50.000 AND
City = 'New York';
```

Id	First-name	last-name	department	salary	City
1	John	Doe	IT	55000	New York
4	Sarah	Brown	IT	53000	New York

11.
```
SELECT first-name,
       last-name,
       department,
       salary
FROM   employees
WHERE  Department IN ('Finance', 'Marketing') AND salary
 > 52.000
ORDER BY salary Desc;
```

id	first-name	last-name	department	salary
3	Mike	Johnson	Finance	60 000
~~9~~	~~Daniel~~	~~Clark~~	~~Marketing~~	~~53 000~~
7	Robert	Wilson	Finance	59000
9	Daniel	Clark	Marketing	53000
5	David	White	marketing	52000

12.
```
SELECT DISTINCT city
FROM employees
WHERE department NOT IN ('IT', 'HR');
```

city
Los Angeles
San Francisco
Houston
Chicago

13.
```
SELECT first_name,
       last_name,
FROM employees
WHERE department != 'Finance' AND salary > 50000
ORDER BY hire_date ASC;
```

first_name	last_name
Emily	Davis
David	White
Jessica	Moore
John	Doe
Sarah	Brown
Daniel	Clark

14.
```
SELECT *
FROM employees
WHERE city IN ('Chicago', 'Los Angeles')
AND department IN ('IT', 'Marketing')
LIMIT 3;
```

id	first_name	last_name	department	salary	hire_date	city
6	Emily	Davis	IT	62000	2015-02-14	Chicago
9	Daniel	Clark	Marketing	53000	2022-06-01	Chicago